

Preethi Vijaykumar Hiremath

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TECHNICAL SKILLS

Backend/DB	Java (Spring Boot, JUnit, Mockito), Python, TypeScript, SQL Oracle (Exadata, PL/SQL, GTT), MySQL, MongoDB, Redis, KDB+
Cloud/DevOps	GCP (Cloud Run), AWS, Docker, Kubernetes, Jenkins, TeamCity
AI/LLM	LangChain, RAG pipelines, Gemini 2.5 Flash, TensorFlow, Keras, scikit-learn
Frontend/APIs	React.js, REST/SOAP, FastAPI, Node.js, Flask, Bootstrap
Security & OSS	AST taint analysis, Linux (RHEL 9, SUSE) PyPI: BrainSurf; VS Code: SnitchLint

EXPERIENCE

Senior Analyst (Software Engineer) Deutsche India Pvt Ltd – Deutsche Bank	October 2024 – March 2026
– Migrated legacy Java/C++ backends to Spring Boot microservices, reducing memory footprint by 95% and eliminating race conditions under 10k+ concurrent users across multi-region traffic – Enhanced real-time Oracle AQ event-driven messaging system delivering sub-second UI updates with 99.9% uptime and zero-drift reconciliation across 6+ interconnected systems – Built ML-driven document analysis pipeline: containerized Python services on GCP (Cloud Run, Docker), integrated Gemini 2.5 Flash via system-prompt engineering to extract financial profiles and generate structured recommendations – Technical leadership of end-to-end CI/CD transformation (Jenkins → TeamCity) across SUSE and RHEL 9; resolved SSL/shared-library conflicts across 90+ components; TDD/BDD cut defect escape rate 30% – Integrated AI-assisted development workflows (GitHub Copilot) into daily engineering; used Cursor with LLM context for complex AST logic in SnitchLint, reducing iteration cycles significantly	
Graduate Analyst (Software Engineer) Deutsche India Pvt Ltd – Deutsche Bank	July 2023 – October 2024
– Designed ETL pipelines integrating 6 upstream/downstream systems with custom transformation and reconciliation logic; thread-safe concurrent processing at 50k+ transactions/hour – Optimized Oracle SQL (GTT, stored procedures, triggers) improving query execution 20%+ across 12 high-volume services; reduced latency 15% during Oracle Exadata migration via schema tuning – Led TLS/SSL migration impact analysis enabling 10+ teams to transition to secure LDAP and Direct connections	
Web Development Intern Luxeveda Brand Services Ltd	April 2023 – June 2023
– Built Node.js backend with MQTT messaging for 50+ IoT devices achieving sub-100ms ingestion latency; designed REST APIs for device scheduling, remote control, and AQI analytics dashboards	
React Developer Intern Inovact Pvt Ltd	August 2021 – November 2021
– Built 15+ responsive React.js components using hooks and state management; reduced page load 20% via code-splitting and lazy-loading; deployed on AWS EC2	

PROJECTS AND RESEARCH

SnitchLint Open-Source VS Code Security Extension TypeScript, AST, Ollama, VS Code LM API	June 2025 – Present
– Published VS Code extension detecting 16+ OWASP vulnerabilities (XSS, SQLi, weak crypto, secrets) with CWE mapping; monotonic fixpoint taint analysis engine with cross-function propagation and source→sink dataflow tracking at ~10ms/500 LOC – Single-pass AST rule pipeline with shared analysis context; optional AI-powered explanations (Ollama/VS Code LM API)	
Trace RAG System for Trade-Settlement Production Support LangChain, Chroma, FastAPI, React, Gemini, MLflow	2026
– Built a RAG system for production-support Q&A over synthetic runbooks and Java code, with dual-source ingestion and ingest-time confidentiality filtering before embedding – Implemented two-stage retrieval (bi-encoder search → cross-encoder reranking) with a tuned similarity threshold, improving relevance over flat vector search – Exposed a FastAPI endpoint on an LCEL chain with a custom Pydantic reranking retriever, decoupled React frontend, and MLflow instrumentation for observability	
BrainSurf EEG Signal Processing Library Python, TensorFlow, Keras, scikit-learn, Streamlit	Jan – July 2023
– Published open-source PyPI library for EEG preprocessing and cognitive state classification using ICA/Wavelet artifact removal and a cross-validated binary classifier (80%+ accuracy)	

EDUCATION

M.Sc. Informatik (Intelligent Systems & Software Engineering) RPTU Kaiserslautern	April 2026 – Present
B.E. Information Science and Engineering B.M.S College of Engineering, Bangalore GPA: 9.19/10.00	August 2019 – July 2023

AWARDS, CERTIFICATIONS AND HACKATHON

Individual Excellence Award Deutsche India Pvt Ltd (Deutsche Bank)	March 2025
Prompt Engineering for Developers Deutsche Bank × Google Cloud Gen AI Program	2025
Financial Inclusion Hackathon Deutsche Bank 2025	
– Built end-to-end ML application in 24 hours: containerized services on GCP (Docker), integrated Gemini 2.5 Flash with system-prompt engineering for document analysis and personalized recommendation generation	